



Identifying restaurant satisfiers and dissatisfiers: Suggestions from online reviews

Anil Bilgihan^a, Soobin Seo^b and Jihee Choi^c

^aFlorida Atlantic University, Boca Raton, FL, USA; ^bHospitality Business Management, Carson College of Business, Washington State University, WA, USA; ^cHospitality Management, The Ohio State University, Columbus, OH, USA

ABSTRACT

Analyzing data from user-generated websites with data mining practices can suggest meaningful insights into services performances. Such techniques may help identifying the precursors of service satisfiers and dis-satisfiers of restaurants. The aim of this research is to analyze online customer reviews for restaurants and underpin the restaurant satisfiers and nature of online restaurant reviews. After retrieving 2214 usable customer comments on Yelp.com, using a coded web spider, a series of both qualitative (e.g., tag clouds, word networks, and word trees were created using data visualization techniques) and quantitative (e.g., MANOVA) data analysis techniques were deployed to detect the satisfiers and dis-satisfiers of different types of restaurants. Three distinct types of clues – functional, mechanic, and humanic – explain a customer's overall perception of a dining experience that lead to satisfaction or dissatisfaction. A focus on food, menu offerings, ambiance, and service create buzz in social media. Furthermore, several implications inform customer researchers about effective ways to investigate and serve the restaurant industry.

KEYWORDS

Online reviews; word-of-mouth (WOM); restaurant industry; service satisfiers; service performance; data mining

使用数据挖掘实践分析来自用户生成的网站的数据，可以对服务性能提出有意义的见解。这样的技术可以帮助识别服务的满意与不满意餐馆的前兆。本研究目的是分析餐厅在线客户评论和支持餐厅满意和网上餐厅自然评论。检索2214可用客户评论Yelp.com后，使用编码的蜘蛛，一系列定性（例如，标签云，字与字树网络，利用数据可视化技术创建的）和定量（例如，MANOVA）数据分析技术被部署到检测的满意和不满意的不同类型的餐馆。线索-功能、机械三种不同的类型，和人性化的-解释一个餐厅的经验，导致满意或不满意的顾客的整体感知。对食品、菜单、气氛和服务的重点，建立在社会化媒体的嗡嗡声。此外，一些影响告诉客户研究人员调查和服务餐饮业的有效途径。

Introduction

The restaurant industry is a key factor in the U.S. economy with respect to its size and its contribution to job creation (Kim, Li, & Brymer, 2016). Food and beverage sales of the restaurant industry in the U.S. reached 745.61 billion U.S. dollars and this figure has been increasing since 1970s (NRA, 2016a). This industry employs 14.4 million

CONTACT Anil Bilgihan  abilgihan@fau.edu  Florida Atlantic University, Boca Raton, Florida.

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employees which translates to the 10% of the total U.S. workforce. Despite the growth, restaurant owners are facing various challenges (Parsa, Self, Njite, & King, 2005). For example, the typical restaurant customer today is not the same as the typical restaurant customer 20 years ago (NRA, 2016b). Younger generations have a very sophisticated world-view when it comes to food compared to older cohorts. Customers have higher expectations of their dining experience and pay more attention to everything from diet-specific food, to sustainability, to food sourcing and production than even just a couple of years ago. To overcome the challenges, it is critical to understand what makes customers choose, return or not return to a restaurant, what makes them recommend a restaurant to their friends and relatives or not recommend it, what image a restaurant portray to its customers, and what features create value for its customers.

An emerging dependence on the Internet as the source of information for decision-making regarding service providers strengthens the need for more research in the electronic reviews area (Berezina, Bilgihan, Cobanoglu, & Okumus, 2016; Kwok & Xie, 2016). Investigating online reviews may help the restaurant practitioners to overcome the challenges they are facing. It is critical for managers to use online customer review information to better understand their customers and improve their performances.

Word-of-mouth communication (WOM) involves the passing of information between a non-commercial communicator and a receiver concerning a brand, a product, or a service, shaping customer behaviors and attitudes (Buttle, 1998; Dichter, 1966). When WOM is mediated through electronic means, the resulting electronic word of mouth (eWOM) refers to any statement that customers share via the Internet (e.g., websites, blogs, micro blogs, social networks, instant messages, online communities) about a product, service, brand, or company (Hennig-Thurau, Gwinner, Walsh, & Gremler 2004; Kietzmann & Canhoto, 2013). People rely on eWOM to make various decisions, ranging from what stocks to invest in to where to dine (Lu, Ba, Huang, & Feng, 2013). The role of eWOM in the hospitality industry is important due to the experience and intangible nature of the hospitality products/services (Jeong & Jang, 2011). With the increasing availability and popularity of Internet opinion platforms, eWOM is an emerging market phenomenon that is playing an important role in consumer decision making process (Liu & Park, 2015). Hicks et al. (2012) examine the motives behind why the people utilize Yelp.com for restaurant reviews. They revealed individuals use Yelp.com for information-seeking more than any other motive, followed by entertainment and convenience. Customers are likely to rely on eWOM that come from experienced customers (Steffes & Burgee, 2009). Hence, a number of recent hospitality research investigated eWOM especially online reviews commenting about restaurants, hotels, and tourism products (Vermeulen & Seegers, 2009; Ye, Law, Gu, & Chen, 2011; Zhang, Ye, Law, & Li, 2010) suggesting significant impacts of eWOM on firm's performance and customers' purchase and visit intentions.

One popular form of eWOM is online reviews. Online customer reviews are defined as "peer-generated product evaluations posted on company or third-party websites" (Mudambi & Schuff, 2010, p.186). Online customer reviews become one of the most effective ways to express customers' thoughts, opinions, and feelings about the products and services (Browning, So, Sparks, 2013; Mauri & Minazzi, 2013; Schindler & Bickart, 2005). The social media environment offers an innovative venue for sharing and gathering

reliable information such as e-Commerce websites (e.g., Amazon, Expedia) and information gathering websites (e.g., Yelp, Tripadvisor). Those websites allow customers to share their experiences and opinions about products/services and provide rating systems to evaluate the products/services using a 5-point scale (1 = unsatisfactory to 5 = very satisfactory). The use of rating system was initiated by Amazon.com in the late 1990s and has become a dominant tool widely utilized by travel websites (e.g., Travelocity) and restaurant review websites (e.g., Yelp).

The exchange of positive and negative information about a business' products and services has a considerable impact on its success. Haywood (1989) suggests restaurant managers to actively listen to WOM and take appropriate actions based on what customers are saying about their business. Monitoring online reviews help marketers to develop new products or services and advertise new or existing offerings to targeted customers (Chung & Buhalis, 2008). Restaurant owners can gain a better understanding of what customers want by investigating the eWOM posted on review websites (Kwok & Yu, 2013). However, these contain a large number of reviews that are difficult and time consuming to wade through and digest. Restaurant review websites, along with a variety of online information sources in the hospitality industry, have become more important given the customers' increasing tendency to share their dining experiences (Tussyadiah & Fesenmaier, 2007). Furthermore, within the restaurant industry, a one-star increase in Yelp rating leads to a 5–9 percent increase in revenue (Luca, 2011). Thus, it is also important for restaurant industry practitioners to grasp what makes their customers satisfied.

Yet, most research to date has focused on the tourism or hotel industry rather than on restaurants, and studies on customer review websites for restaurants are relatively limited (Jeong & Jeon, 2008; Levy, Duan, & Boo, 2013), demonstrating a need for the current research. Most of the research regarding online review sites also relied on only one type of research method approach, which might not be sufficient to have holistic view on customer's behavior in online restaurant review context. Extant hospitality studies investigated the satisfiers and dissatisfiers of restaurants using self-administrated surveys; however, extensive customer feedback and reviews online are an unmeasured and untapped wealth of information about customer satisfaction (Pizam, Shapoval, & Ellis, 2016).

The primary aim of the present study is to provide insight into how customers share the restaurant-related information and/or experiences in online review websites by using a mixed-method (i.e., qualitative and quantitative) approach. Thus, the objectives of this research are to (1) *analyze online reviews for restaurants using data mining practices to investigate the restaurant satisfiers/dis-satisfiers* and (2) *identify the factors influencing the restaurant ratings in online review websites*. Multiple quantitative and qualitative techniques are used to help restaurant operators point to areas in need of improvement and achieve a better understanding of what customers want and how they perceive the restaurant. Visualization techniques were used, which supports exploration, discovery, and complex reasoning about data and data-driven problems (Ribarsky, Wang, & Dou, 2014), offering useful guidelines for restaurants wishing to enhance their customers' satisfaction and achieve high ratings online. As one of the predominant restaurant online review websites, the current study focused on Yelp.com.

In sum, this study proposes and aims to answer two research questions:

- What are satisfiers/dis-satisfiers in online review on Yelp.com?
- What are factors influencing online ratings of restaurants on Yelp.com?

Literature review

Online customer reviews and yelp.com

Customers are using the Internet to acquire information about products and services, and the widespread adoption of Web 2.0 plays a crucial role in the development of eWOM (Chan & Guillet, 2011). Compared to traditional WOM, eWOM is easy and cost-efficient and spreads to more people in a shorter period of time, which spreads usually by email, online reviews, and blogs (Litvin, Goldsmith, & Pan, 2008). Among those media, online reviews are a form of eWOM which is defined as user-generated product or service evaluations that are posted on third party websites or company websites (Mudambi & Schuff, 2010). Online reviews are considered as powerful sources of information in customers' decision making (Ye, Law, Gu, & Chen, 2011; Filieri & McLeay, 2014) because customers share their hostility industry experiences with other customers via the Internet (Park & Allen, 2013). They significantly influence sales, attitudinal change, and firm's reputation (Vermeulen & Seegers, 2009; Sparks & Browning, 2011; Ye, Law, & Gu, 2009). There are various types of online customer review sites such as e-commerce and product review sites (e.g., Amazon.com), restaurant review sites (e.g., Yelp.com), travel review sites (e.g., Tripadvisor.com), which enables the rapid and extensive distribution of comments and offers individuals the ability to share information. The role of online reviews, as one popular form of social media, is particularly important for the hospitality and tourism industries (Zhang et al., 2010) because these industries are intangible and consist primarily of an experiential component. Therefore, customers seek online reviews from experienced sources to reduce uncertainty and risks associated with the service (Berlyne, 1966; Xiang & Gretzel, 2010; Zhang et al., 2010). When customers need to plan a trip, or make a hotel or restaurant reservation, they rely on online reviews to choose a service provider (Jeong & Jang, 2011; Kim, Mattila, & Baloglu, 2011). Customers tend to trust other customers' reviews since they are perceived as relatively unbiased, reliable and independent of a company's direct marketing efforts; accordingly, they may exert more impact on customers' choices (Chen & Xie, 2008; Li & Bernoff, 2008).

The increasing importance of online reviews in the hospitality and tourism industries has encouraged academic analysis in various contexts. A pioneering work of Pantelidis (2010) performed a content analysis for 2,471 restaurant reviews in the U.K., revealing the most frequently appeared words and menu items in customer comments to identify the salient features influencing restaurant experience. Another study by Kwok and Yu (2013) using a text-mining technique examined 982 Facebook comments to investigate the business- customer communication through social media, providing most indicative words appeared in the social media communication.

A number of online review websites like Yelp, Amazon, and OpenTable, allow users to rate the level of satisfaction about the service or products based on their experiences. For example, Yelp.com provides the overall rating service by asking users to indicate the level of their experience using a five-star rating scale from five stars to one star. Yelp is a San

Francisco based Web 2.0 website where customers can read and post reviews for restaurants and other various businesses such as hair stylists, shops, spas and mechanics. Yelp was founded in 2004 to help customers find local businesses. The website had an average of approximately 130 million visitors per month and it hosts over 36 million local reviews. It creates a virtual community where users can create profiles with avatars and discuss various topics (e.g., best restaurants/bars/coffee shops in a location) with other users. Business owners or managers can also setup an account to post photos and messages. The website also uses an automated filter that suppresses suspicious reviews. Users can search for specific restaurant reviews or filter the restaurants based on location, price range, and food type. On Yelp, users can post comments about their experiences with restaurants. Yelp also holds the information about the restaurants, including address, business hours, whether the restaurant accepts credit cards or not; takes reservations; delivery options; attire; parking options; alcohol options; WiFi; noise level; good for (e.g., breakfast, lunch, dinner, late night); ambiance; catering options; wheelchair access. Figure 1 represents an example of restaurant profile on Yelp.

Online review rating systems are considered as reliable and they influence customers' decision making in the hospitality industry (Cheng & Loi, 2014; Filieri & McLeay, 2013; Liu & Park, 2015; Park & Nicolau, 2015; Serra Cantallops & Salvi, 2014). Several studies (e.g., Ye et al., 2009, 2011; Zhang et al., 2010) found the positive relationship between online customer ratings and purchasing behavior, which they used as one of proxies for financial performance. Xiang, Schwartz, and Gerdes, & Uysal (2015) highlighted it is essential for companies to understand the antecedents leading to customers' satisfaction because overall ratings impact companies' financial performance. It is important to identify the characteristics associated with online customers' satisfaction (Radojevic, Stanistic, & Stanic, 2015). Furthermore, Casaló Flavián, Guinalú and Ekinci (2015) investigated the influence of online hotel rating using an experimental study, asserting that the online rating is more persuasive when the rating is provided by well-known



Figure 1. An example of restaurant profile on yelp.com.

review websites such as TripAdvisor than not well-known websites. Therefore, it is valuable to investigate what factors affect customers' satisfaction (high rating) or dissatisfaction (low rating) in well-known online review sites.

Despite a number of research on social media and online reviews, lack of attention has been paid to which factors that drive the overall restaurant ratings. Furthermore, compared to the lodging industry, there is a lack of studies that identify online customer reviews as a key determinant of restaurants' performance due to the challenge of collecting online reviews (Kim et al., 2016). Particularly, it is because small and medium size restaurant units rarely have the luxury to access the data or fund personnel to trace customers' online review information. Whereas, hotel managers receive rich online review data from corporate and manage online reviews in order to enhance their overall customer experience.

Thus, this study analyzes online customer reviews to identify satisfiers and dis-satisfiers of restaurants using visual data mining technique as well as examine factors influencing the online ratings of restaurants on Yelp.com. The primary goal of this study is to provide insights into restauranters and food and beverage researchers. Multiple quantitative and qualitative techniques are used to help restaurant operators point to areas in need of improvement and achieve a better understanding of what customers want and how they perceive the restaurant. Visualization techniques were used, which supports exploration, discovery, and complex reasoning about data and data-driven problems (Ribarsky et al., 2014), offering useful guidelines for restaurants wishing to enhance their customers' satisfaction and achieve high ratings online. As one of the predominant restaurant online review websites, the current study focused on Yelp.com.

Theoretical framework (herzberg's motivator and hygiene factor theory)

Herzberg's Motivator and Hygiene Factor Theory (Herzberg, Mausner, & Snyderman, 1959) has been suggested as applicable to investigate dimensionalities of customer satisfaction (Oh & Park 1997). Herzberg's theory, initially created for job satisfaction, classifies two categories of forces that are mutually exclusive or individually independent and affect behavior in different ways. The two categories of forces are: a) hygiene/dis-satisfiers and b) motivator/satisfiers. According to this theory satisfaction and dissatisfaction are not two opposite extremes. The theory further argues that the absence of hygiene factors leads to dissatisfaction, therefore dis-satisfiers are attributes that contribute to dissatisfaction when performance is low but do not enhance satisfaction when performance is high as the customers take these factors for granted (Füller, Matzler, & Faullant, 2006). Satisfiers on the other hand, do not necessarily lead to dissatisfaction if they are not delivered. Hence, identifying both satisfiers and dis-satisfiers are crucial to understand factors influencing customer satisfaction.

Online customer reviews reflect customer satisfaction/dissatisfaction about customers' experience with products/services. Customer satisfaction is a crucial indicator of a company's past, present, and future performance (Oliver, 1999) and directly reinforces post-purchase behaviors such as attitude change, repeat purchase, and brand loyalty (Fornell, 1992). On the other hand, dissatisfied customers tend to complain to the establishment or seek redress from them more frequently, perhaps to relieve cognitive dissonance and compensate for a failed consumption experience (Oliver, 1987). In particular, both satisfied and unsatisfied customers are likely to leave comments or share their experiences through the electronic reviews (Bansal & Voyer, 2000; Gremler, Bitner, & Evans, 1994;

Litvin et al., 2008). Negative comments can also be an opportunity for improvement of service quality if managed properly. Kandampully, Zhang, and Bilgihan (2015) suggest that with the growth of social media, online forums, virtual communities and C2C exchange platforms, companies can learn more about their customers. Hence, companies need to closely monitor social media to re-design their services based on online customer comments to ensure customer satisfaction and loyalty.

Visual data mining techniques

Given the number of online reviews, data mining techniques have emerged as powerful tools to analyze online customer reviews. Especially, visual data mining techniques have high value in exploratory data analysis, as well as great potential in terms of exploring large amounts of data, especially when little is known about the data or the exploration goals are vague (Keim, 2002; Viégas et al., 2007). Not only are interactive visualizations a key medium for communication in a data-rich world, but preliminary reports suggest that visualizations may catalyze storytelling (Viégas, Wattenberg, & Dave, 2004) and collective data analysis (Wattenberg, 2005), which is especially fitting for studying experience-based products such as dining experiences. Visualization techniques are used to help identifying the common themes by analyzing narrative reviews.

There are a few computer scientists employing data mining techniques to analyze restaurant online reviews. Ye et al. (2009) employed sentiment analysis of online reviews using machine learning algorithms to identify the valence of restaurant online reviews (positive/negative), and similarly Zhang et al. (2011) used the sentiment analysis for investigating restaurant online reviews that are written in Cantonese. Another group of scholars examined the topics, keywords, and word patterns of online reviews using data mining techniques. For instance, Lee and Hu (2004) examined the most frequently appeared words from negative customer feedback, Capriello, Mason, Davis and Crotts. (2013) investigated word patterns from vast amount of travel reviews using a semantic clustering method.

The visual data mining techniques are also used in the hospitality and tourism research to identify the airport service quality drivers of passenger satisfaction (Bogicevic, Yang, Bilgihan, & Bujisic, 2013), hotel service drivers of customer satisfaction (Bilgihan, Severt, & Kandampully, 2013), and patient experience (Maramba et al., 2015), while no research has been conducted in the context of restaurant experiences. The visual data mining techniques are expected to deliver useful guidelines for restaurateurs by identifying satisfiers and dis-satisfiers featured in online reviews.

Method

Data collection: the use of web spider

This research gathered and analyzed data from Yelp using various software including SPSS, Linguistic Inquiry and Word Count (LIWC) and ManyEyes.

¹Quantcast is a technology company that offers public access to traffic and demographic data for websites and detailed user insights.

Table 1. Demographic profile of yelp.com visitors.

Gender		Age		Children in household	
Male	45%	<18	6%	No Kids	63%
Female	55%	18–24	16%	Has Kids	37%
		25–35	26%		
Education Level		35–44	25%	Household Income	
No College	35%	45–54	16%	\$0–50k	14%
College	45%	55–64	8%	\$50–100k	22%
Grad School	20%	65+	4%	\$100–150k	28%
				\$150k+	36%

According to the report of quantcast¹.com on the demographic profile of Yelp.com users, the users of Yelp.com consist of 45% males and 55% females. Table 1 displays the US demographics for Yelp (generated from quantcast.com).

In order to collect data (data includes restaurant profiles and customer comments) from Yelp, a web spider is deployed. A web spider, crawler, or robot is a computer application that iteratively and automatically downloads web pages (Thelwall, 2001). For this research, the application recursively visits the target website (i.e., www.yelp.com) and extracts the content of interest. Using a web spider is a very effective method for collecting information from websites that include user-generated content (Gerdes, Jr & Stringam, 2008). The data collection process included both quantitative and qualitative data collection. Restaurant experiences provide the unit of analysis for this study, and the focus was randomly selected restaurants in the United States. Qualitative customer comments were intended to use mainly for exploratory purposes by presenting data visualizations. Further qualitative analysis included LIWC (Pennebaker, Francis, & Booth, 2001), word frequencies, and content analysis. The quantitative analysis includes descriptive statistics, and multivariate analysis of variance (MANOVA) in order to categorize the effect of restaurant characteristics on the overall rating of restaurants.

Sample selection

The web spider randomly selected customer comments posted over a period of two years on www.Yelp.com. This process resulted in 2214 responses, which is comparable to the sample sizes of text mining studies. For example, Na, Thet, and Khoo (2010) used a sample of 520 online movie reviews to conduct sentiment analysis. White (2013) used a sample of 130 to use data visualization tools. Data was saved in a CSV file, the first row included the variable names. Positive and negative recommendations could be identified on the basis of the overall satisfaction score; the quantitative scores ranged from 1 (most negative valence) to 5 (most favorable valence). Collected data included qualitative reviews of the restaurants.

Visualization pattern analysis

This study employed visualization pattern analysis using Many Eyes by IBM application which enables the power of human visual intelligence to find patterns. The aim of this analysis is to generate visualization patterns used written comments associated with point scores to organize the data. It draws networks of words, such that two words are connected if they are associated in qualitative comments, because they appear together

in a phrase of the form “X and Y.” Common words such as “the” or “of” are excluded, because they are not informative for this kind of display. This method allows researcher to recognize and identify common emerging themes. A word tree offers a visual search tool for unstructured text, such as a book, article, speech, or poem, that allows readers to pick a word or phrase and then indicate all the different contexts in which that word or phrase appears. The contexts get arranged in a tree-like branching structure to reveal recurrent themes and phrases. The word tree starts with a blank slate instead of full data visualizations; the researcher then chooses a search term to display. The software finds all occurrences of that term, along with the phrases that appear after it. Tag clouds were deployed to see the overview of the most salient terms in a large text corpus (Viégas, Wattenberg, Van Ham, Kriss, & McKeon 2007). The sections below explain the different visualization analysis techniques deployed in this study.

The word tree

The “Word Tree” is a visualization and information retrieval technique that is used to analyze text documents (Wattenberg & Viégas, 2008). A word tree is a graphical form of the traditional “keyword-in-context” (KWIC) method, and enables rapid querying and exploration of bodies of text. It is an interactive form of KWIC method. It shows the words that follow the user query. For example, if the user searches the term “service” in a text file of customer reviews, it shows the “service” as the root node, then the various distinct subsequent word define branches. User can click on child nodes to explore further. For instance, user can click on “is” to create a child node of “service is” to explore the text further.

Phrase nets

A phrase net maps the relationships between different words used in a text. It is a technique for mapping unstructured text files. This method involves two steps. First is a pattern-matching phase, in which the linkages are defined between words based on user-defined templates. The second phase involves laying out and rendering the graph. The layout phase includes a new, general clustering procedure that reduces the complexity of the graph while retaining its information content (Van Ham, Wattenberg, & Viégas, 2009). It uses pattern-matching techniques to provide multiple views of the concepts contained in a text document. It analyzes the text document by looking for pairs of words that fit particular patterns. The size of a word is comparative to its occurrence in a match; the width of an arrow between words is associated with how many times those two words occur in the same phrase. Finally, the color of a word specifies whether it is more likely to be found in the first or second slot of a pattern. The darker the word, the more often it appears in the first position.

Heatmaps

A heatmap is a two-dimensional graphical illustration of data that shows the values of a variable as colors. They display the data directly over the stimulus. Heatmaps show the data close to the elements to which the data pertains, therefore, it is simple to read a heatmap. They can be of great value to aericles, reports, and presentations because they summarize large quantities of data that would be much more difficult to grasp if presented numerically (Bojko, 2009). Heatmaps help researchers quickly see “the big picture” including any patterns or trends that may exist in the data.

Linguistic inquiry and word count text analysis

LIWC text analysis is a measure of emotion expressions (Kahn, Tobin, Massey, & Anderson, 2007). It calculates the degree to which people use different categories of words across a wide array of texts. LIWC analyzes text samples and compares words to a dictionary. Categories such as positive emotion words, were formed initially by having independent judges decide which words should go into each category (Newman, Groom, Handelman, & Pennebaker, 2008). Therefore, human judgment is combined with an automated and consistent coding system. LIWC recognizes about 80% of the words in a given text sample (Newman et al., 2008). LIWC method allows the user to enter text and it investigates this text in respect to associated words for seven dimensions, namely self-references, social worlds, positive emotions, negative emotions, overall cognitive words, articles and big words. Each dimension is linked to by some target words from a pre-defined dictionary. This method was previously used in online contexts such as MySpace (Pfeil, Arjan, & Zaphiris, 2009), gender differences in language use (Newman et al., 2008), students' daily conversations (Mehl & Pennebaker, 2003), and Internet communications (Owen, Yarbrough, Vaga, & Tucker, 2003).

Data analysis

The analysis of data consists of two folds: quantitative analysis and qualitative analysis. The qualitative analysis involves aforementioned methods, namely, word frequencies, word clouds, word trees, LIWC text analysis, and network maps. The quantitative analyses involve descriptive statistics, and multivariate analysis of variance (MANOVA) using SPSS. The MANOVA is utilized to identify the effect of restaurant characteristics (e.g., waiter service, parking, outdoor seating, good for kids, good for when) on the overall rating of restaurants. The independent variables were mostly categorical variables, not continuous variables (e.g., WiFi, delivery, outdoor seating) that can be coded as 0 or 1, and dependent variable was the overall rating. In addition, the data is divided into two groups based on the overall price (high vs. low priced groups) for multi-group analysis to investigate whether significant factors differ depending on the price level of the restaurants.

Results

Qualitative results

After collecting data from Yelp, for the qualitative part of the study, the next step was to divide the dataset into two separate sets based on the overall scores to identify the factors that are satisfiers for restaurants based on Hygiene Factor Theory. Andaleeb and Conway (2006) highlight the importance of identifying the satisfiers in the service industry. Particularly, it is essential to understand the dynamics of the service industry from the customers' perspective, who is the final arbiter of where, when and what to eat. Thus, a comprehension of the factors that influence the customer satisfaction should be useful in advising restaurateurs or managers to design and deliver the right offering. For the qualitative analysis, we split the dataset into two parts by sorting the ratings using SPSS. With a purpose of determining the factors that explain customer satisfaction using online customer reviews, the comments with overall high ratings (4 and 5 out of 5-point scale)

are included, with the assumption of high overall ratings mirror customer satisfaction. This process yielded a sample size of 1294. Later, the positive comments were analyzed using IBM ManyEyes.

The most frequently occurred words for satisfied customers included *food*, *good*, *place*, and *great*. Other highly mentioned words included *delicious*, *restaurant*, *like*, *service*, *best*, *menu*, *order*, *little*, and *pretty*. Based on word clouds and word frequency counts of positive review, it could be argued that a focus on food, menu offerings, ambiance, and service creates buzz in social media.

Word-trees and phrase nets

To investigate the underpinnings of the satisfiers, a series of word-trees were deployed. Word trees reveal a branching view of how revisers used the key words and phrases. For instance, using the most occurring words for satisfied customers, “*food*” was chosen (847 mentions) to start a word tree. After “*food*”, “*is*”, “*was*”, and “*and*” were the most occurring phrases (refer to Figure 2 for an example of word tree).

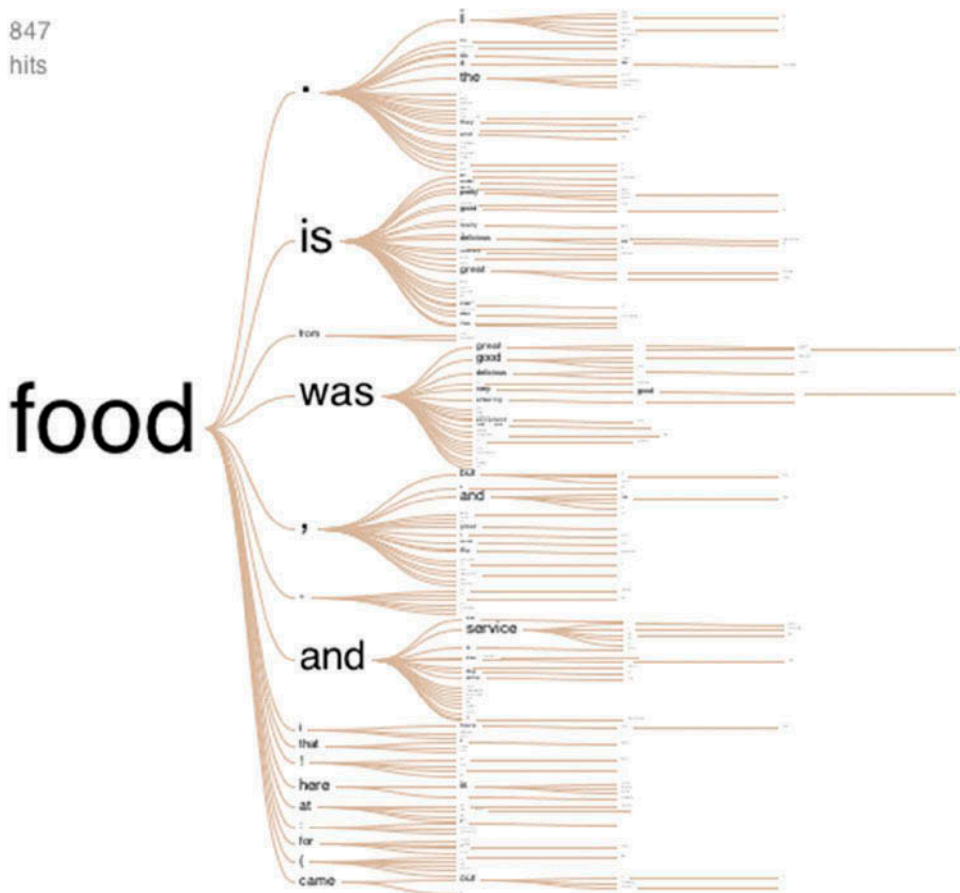


Figure 2. Word tree of “Food”.

Following, another word tree using “food was” deployed. Word tree of “food was” brings to light that great, excellent, amazing and outstanding were the most used words after “food was”. Word trees for food related phrases revealed that reviewers tend to comment about hedonic features of the food. With such methodology, in-depth factors were identified. For example, a word tree of “server” was chosen to pinpoint the see what customers were saying about servers. The word tree shows *knowledgeable*, *friendly*, *informative*, *attentive*, *fantastic*, *awesome*, *pleasant*, and *helpful* were the attributes emerged under “server” word tree. The word “service” was mentioned 403 times, therefore, a word tree of “service” was deployed and it was found that *great*, *friendly*, *good*, *prompt*, and *excellent* were the characteristics emerged under “service”. Further word trees were deployed to assist the process of content analysis.

The next step in analyzing the qualitative reviews sought to discover networks of related words. Figure 3 represents the networks of related words contained in four and five star reviews, including the strongest relationship between “food” and “service”. “Fresh” is related to “delicious”, “delicious” is linked with “flavorful”, and “flavorful” is related to “tasty” and “rich”. “Light” is related to “flavorful”, “tasty”, and “delicious”. Such strong relationships indicate the importance of food quality. “Prompt”, “friendly”, “attentive”, “helpful”, “nice”, “warm” were all related in word networks which signifies the importance of staff and service quality.

Similarly, a phrase net diagram for negative reviews ($n = 338$) was created. Not surprisingly, it was found that service and food were the main reasons of service failure in restaurants. However, word frequencies also show a clear pattern that service components are the main antecedents of negative reviews. People tend to not to complain about the food as much as they do for service related issues. “Slow”, “rude”, “wrong order” were all related with the negative aspects of the service whereas “bland”, “greasy”, “mushy”, “dry”, and “heavy” were related with the food factor.

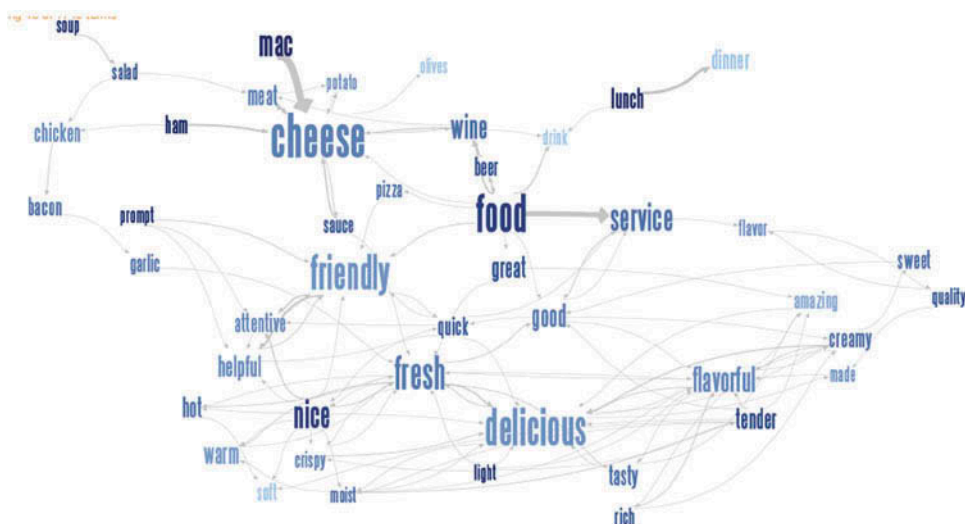


Figure 3. Phrase net of positive reviews.

Content analysis of online reviews

Following, a content analysis was deployed leaning on Berry, Wall, and Carbone (2006)'s experience clues typology. The model claims that customers form perceptions based on three distinct clues namely, functional, mechanic and humanic. Functional clues are the technical performance of the service whereas mechanic clues signify the tangibles associated with the service. Humanic clues are associated with the behavior and appearance of service providers. Those clues play specific roles in creating the customer's service experience, influencing both rational and emotional perceptions of service quality (Berry et al., 2006; Jain & Bagdare, 2009). Wall and Berry (2007) later used this typology to study the multilayered experience of restaurant service quality.

The frequently occurring words were grouped under this model. Functional clues are the core products, the basis of a restaurant's success, therefore they include food and beverage. Under the food factor, *fresh, taste, delicious, flavor, rich, light options, menu variety* and *presentation* emerged. Beverage factor included *local, draught, taste, delicious* and *presentation*. Mechanic clues are nonhuman elements in the service environment such as *equipment, lighting, and color*. *Atmosphere, layout, music, design, outdoor, location, décor, and cleanliness* are also identified. Humanic clues include the behavior of employees. Our analysis yielded two factors of humanic clues, service and experience. Service component is directly related to the behavior of servers whereas experience factor emerged as the humanic clues of customers (themselves and other patrons in the establishment). *Attentive, helpful, greet, friendly, and prompt* emerged in the service factor. *Mood, feel, crowd, and fun* emerged under the experience factor. Figure 4 represents the content analysis of restaurant quality dimensions. *Price* also emerged as another factor in online reviews, since it was not included in the original typology, which is kept as a separate factor that customers mention in their reviews.

Linguistic inquiry and word count text analysis

LIWC text analysis was used as a measure of emotion expressions. LIWC calculates the degree to which people use different categories of words across a wide array of texts. This method allows the user to enter text and it investigates this text in respect to associated words for seven dimensions, namely self-references, social worlds, positive emotions, negative emotions, overall cognitive words, articles and big words. Each dimension is linked to by some target words from a pre-defined dictionary. This method was previously used in online contexts such as MySpace (Pfeil et al., 2009). Table 2 shows the LIWC results in comparison with personal and formal texts libraries.

LIWC analysis revealed that restaurant customers use a high rate of self-references (when compared to formal texts) implying that they tend to be more insecure and nervous. However, self-references in restaurant reviews occur less when compared to personal texts denoting they tend to be less honest than personal texts. Social words in restaurant reviews occurred almost at the same level with other texts, suggesting people refer to other people (e.g., service employees, other customers). This also means that they are outgoing and socially connected (Pfeil et al., 2009). Positive emotions occur more than negative emotions in restaurant reviews. This is mainly due to the fact that Yelp reviews are skewed toward positive reviews. Cognitive words also occurred high signifying that the users are actively thinking about their writing topic. Usage of articles was the highest compared to both personal and formal texts, meaning users tend to be more concrete and

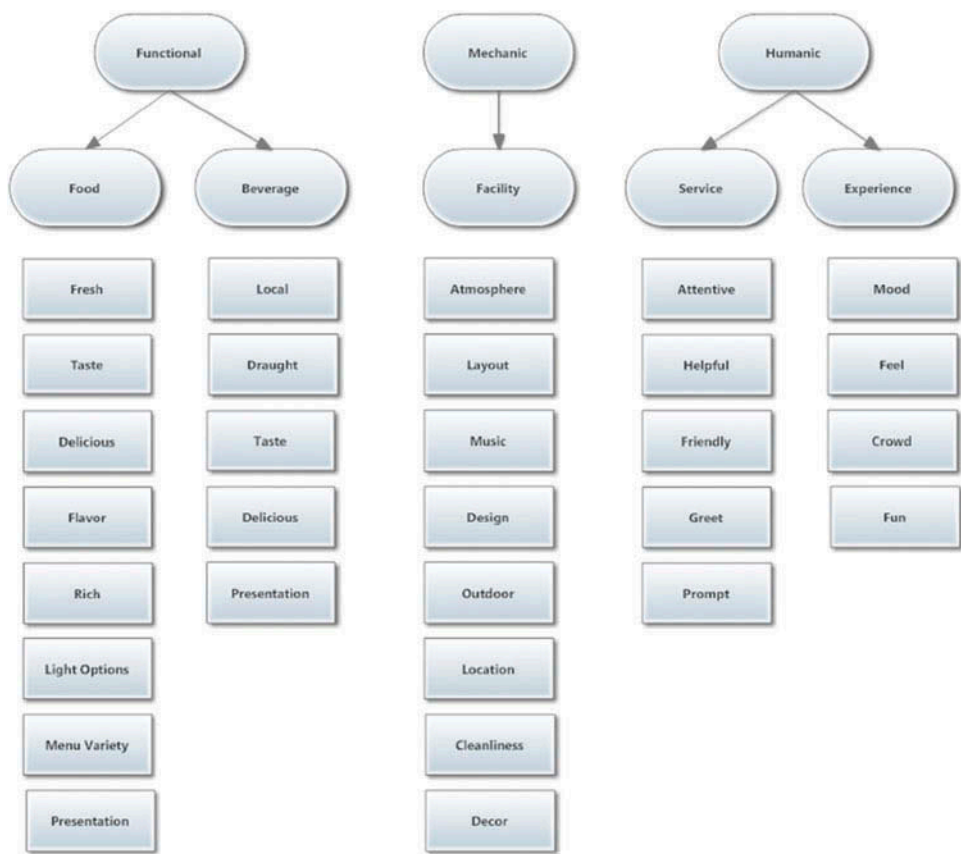


Figure 4. Restaurant quality dimensions.

Table 2. LIWC Results.

LIWC Dimension	Examples of linked words	Yelp data	Personal texts	Formal texts
Self-References	I, we, me	9.36	11.4	4.2
Social Words	Talk, us, friend	9.73	9.5	8.0
Positive Emotions	Happy, good, nice	4.08	2.7	8.0
Negative Emotions	Hate, worthless	0.97	2.6	1.6
Overall Cognitive Words	Cause, know, ought	7.21	7.8	5.4
Articles	an, a, the	8.27	5.0	7.2
Big words	(> 6 letters)	15.64	13.1	19.6

impersonal in their thinking. They also use “big words” compared to personal texts, thus, reviewers are portraying themselves as well-educated. This implies that restaurant reviews are not solely about the restaurant business, but also it is about the reviewer.

Heatmaps

The last step for the qualitative part used heatmaps to discover whether we could cluster the restaurants based on their locations and high occurring words (e.g., are there specific places that customers comment about “breakfast” heavily). Heatmaps

Table 3. MANOVA Results. the effects of restaurant characteristics on the overall ratings of restaurant reviews.

Source		N	Mean	S.D.	F	Sig.
Waiter Service	No	742	3.64	1.34	17.70	.000***
	Yes	1472	3.87	1.11		
Outdoor Seating	No	1083	3.65	1.29	30.17	.000***
	Yes	1131	3.93	1.09		
Delivery	No	1500	3.97	1.07	104.61	.000***
	Yes	714	3.42	1.36		
Wi-Fi	No	1736	3.76	1.25	15.86	.000***
	Yes	103	4.25	.81		
Good For When	Brunch, Lunch	1225	4.05	.99	130.08	.000***
	Dinner, Late night	989	3.48	1.35		
Alcohol	No	747	3.67	1.36	12.38	.000***
	Yes	1467	3.86	1.10		
Good For Kids	No	1191	3.92	1.08	28.51	.000***
	Yes	1023	3.65	1.30		
Accept Credit Cards	No	207	2.37	1.46	375.61	.000***
	Yes	2007	3.94	1.06		
Parking	Street	1265	3.63	1.31	56.82	.000***
	Private lot, Valet	949	4.01	.99		
Good for Groups	Yes	854	3.68	1.34	12.96	.000***
	No	1360	3.87	1.09		
Noise Level	Quiet	83	3.67	1.38	7.47	.001**
	Average	1525	3.79	1.21		
	Loud	84	4.30	.97		
Attire	Dressy	132	3.61	1.30	3.17	.075
	Casual	2082	3.81	1.19		
Has TV	No	1526	3.84	1.21	3.06	.080
	Yes	547	3.73	1.13		
Caters	No	1833	3.80	1.21	.40	.523
	Yes	381	3.76	1.12		
Takes Reservation	No	1278	3.78	1.25	.23	.627
	Yes	936	3.81	1.12		
Take Out	No	74	3.74	1.35	.13	.710
	Yes	2140	3.80	1.19		

*p < .05, ** p < .01, *** p < .001

Table 4. Descriptive statistics and Univariate ANOVA results.

		Low priced groups			High priced groups			ANOVA	
		Mean	SD	N	Mean	SD	N	F	Sig.
Delivery	No	4.15	.92	492	3.88	1.12	1008	174.22	.000***
	Yes	2.72	1.46	275	3.86	1.08	439		
Good for groups	No	2.92	1.49	320	4.13	1.00	534	247.47	.000***
	Yes	4.15	.91	447	3.73	1.15	913		
Outdoor seating	No	3.48	1.38	599	3.86	1.13	484	32.97	.000***
	Yes	4.20	.95	168	3.88	1.10	963		
Caters	No	3.58	1.36	683	3.93	1.09	1150	25.68	.000***
	Yes	4.11	.91	84	3.66	1.15	297		
Alcohol	No alcohol	3.42	1.40	556	4.40	.91	191	114.60	.000***
	Alcohol	4.22	.89	211	3.80	1.12	1256		
Parking	Street parking	3.15	1.49	379	3.83	1.16	886	66.67	.000***
	Private lot, Valet	4.11	.93	388	3.94	1.02	561		
Good for when	Brunch, Lunch, all	4.11	.91	560	4.00	1.05	665	199.67	.000***
	Dinner, Late night	2.37	1.46	207	3.77	1.15	782		

*p < .05, ** p < .01, *** p < .001

show a high density of keywords that appear in reviews. Boston, MA is chosen and provided heatmaps of “*tourist*” and “*pricey*” (see Figure 5) and it is apparent that reviewers mention similar keywords in the same neighborhoods.

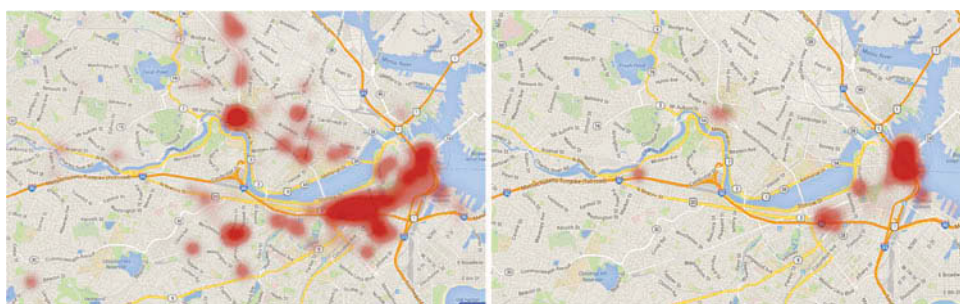


Figure 5. Heatmap of keyword pricey and tourist.

Quantitative results

MANOVA results

The Yelp.com allows customers to rate the restaurants using an indicator, overall rating, which ranged from 1 (poor) to 5 (excellent). The effects of restaurant characteristics on the overall ratings were examined using multivariate analysis of variance (MANOVA). The restaurant characteristics provided by yelp include waiter service, outdoor seating, delivery, Wi-Fi, good for when, alcohol, good for kids, accept credit cards, parking, good for groups, noise level, attire, has TV, caters, take reservation, and take-out.

Table 3 revealed significant factors affecting the overall ratings of restaurant reviews: waiter service, outdoor seating, delivery, Wi-Fi, good for when, alcohol, good for kids, accept credit cards, parking, good for groups, and noise level. Significantly higher overall ratings were detected when a restaurant had waiter service compared to a restaurant without waiter service such as take-out restaurant ($F = 17.70, p < .001$). Restaurants with outdoor seating showed higher ratings than those of restaurants without outdoor seating ($F = 30.17, p < .001$). Restaurants that had delivery service exhibited higher ratings than those of restaurants without delivery service ($F = 104.61, p < .001$). Whether a restaurant had Wi-Fi service influenced the overall ratings, higher ratings were detected for restaurants with Wi-Fi than those without Wi-Fi service ($F = 15.86, p < .001$). When restaurants were good for brunch or lunch rather than dinner or late night, higher overall ratings were found ($F = 130.08, p < .001$). Restaurants that serve alcohol drinks had higher ratings than restaurants that do not serve alcohol drinks ($F = 12.38, p < .001$). When restaurants were perceived to be good for kids, higher ratings were found compared to when restaurants were not good for kids ($F = 28.51, p < .001$). For restaurants that accept credit cards, higher ratings were found opposed to restaurants that do not accept credit cards ($F = 375.61, p < .001$). When restaurants offer private parking lot or valet parking service than street parking, higher ratings were detected ($F = 56.82, p < .001$). When restaurants were perceived to be not good for groups, higher ratings were found compared to restaurants that were good for groups ($F = 12.96, p < .001$). The level of noise was found to influence the overall ratings, highest overall ratings were found for loud restaurants, lowest for quiet restaurants, and average-loud restaurants fell in the middle ($F = 7.47, p < .01$). On the other hand, several restaurant characteristics were found to be insignificant factors influencing overall ratings: attire, whether a restaurant has TV, catering service, takes reservation, and take-out.

Univariate ANOVA results

In order to identify the determinants of high ratings depending on the type of restaurants, sampled data were divided into two groups depending on the average menu price of reviewed restaurants: low priced groups ($n = 767$) and high-priced groups ($n = 1447$). The factor called “price” was examined in its interaction effect with other restaurant characteristics that were found to be significant determinants of high ratings based on the result of MANOVA. Some variables were not applicable to test interaction effects with price such as waiter service, WiFi, good for kids, and accept credit cards due to its skewed nature of variables.

Results of Univariate ANOVA found significant two-way interaction effects between 530price and delivery, good for groups, outdoor seating, caters, alcohol, parking, and good for when. It is found that the delivery service, good for groups, outdoor seating, parking, and good for when are critical factors for low-priced groups while are not critical for high priced groups (Table 4). While there may be a number of restaurant characteristics heavily influencing the ratings of low-priced restaurants, those characteristics may have less impact on the ratings of high-priced restaurants because customers tend to value food quality more than other factors when dining at high-priced restaurants.

For low-priced restaurants, providing delivery service led to lower ratings than one without delivery service. When a low-priced restaurant is considered good for groups, the restaurant had a higher rating compared to a low-priced restaurant not good for groups. A restaurant that provides outdoor seating, catering service, and alcohol products showed higher ratings compared to that of a restaurant without outdoor seating, catering service, and alcohol products. The private parking provided by a restaurant led to a higher rating than street parking. When a restaurant is good for brunch or lunch, the restaurant had a higher rating than a restaurant considered good for dinner.

Discussion, implications, limitations, and future directions

Discussion

This study analyzed online reviews from Yelp.com using multiple visualization techniques (word tree, phase nets, heatmap, and LIWC text analysis) along with quantitative analysis technique to identify satisfiers and dissatisfiers of online reviews. Findings are important for both researchers and practitioners. From a methodological perspective, an innovative data analysis technique for customer reviews was introduced. From a managerial point of view, findings help restaurateurs to understand customer’s needs. Results show how customers perceive restaurants and identify potential areas for improvement. Consequently, practitioners can then use this information to develop new products or services and advertise new or existing offerings. The current study results indicate that the most frequently mentioned word in both positive and negative reviews was “food,” which is unsurprising since food is the primary purpose for dining out. This supports the results of a study by Namkung and Jang (2007), showing that food quality is considered a vital component in restaurant customer satisfaction. Similarly, Baker and Crompton (2000) found that the food quality plays a fundamental role in reinforcing patrons’ behavioral intention toward restaurants. Obviously if the primary product (i.e., food) is ignored, that may taint the customer’s experience and generate negative comments regardless of excellent service or a well-priced menu.

Content analysis of online reviews revealed three distinct types of clues – functional, mechanic, and humanic – that explain a customer's overall perception of the dining experience. Previous studies on the importance of managing service experience clue, concluded that functional, mechanic, and humanic clues play specific roles in creating the customer's service experience, influencing both rational and emotional perceptions of service quality (Berry et al., 2006). In addition, Andaleeb and Conway (2006) find that customer satisfaction was influenced by the responsiveness of frontline employees (a humanic cue), followed by price and food quality (functional clues). However, the restaurant's physical design and appearance (mechanic clues) did not have a significant effect on satisfaction. In terms of providing a satisfying customer experience, it is important for restaurants to design and orchestrate clues correctly.

LIWC text analysis was used to measure reviewers' emotional expression along with seven dimensions: self-reference, social words, positive emotions, negative emotions, overall cognitive words, articles and big words. In Yelp reviews, social words appeared almost as often as other texts, suggesting that people tend to make references to other people (e.g., friends, service employees, other customers) in their online reviews. This also suggests that reviewers are more likely to be outgoing and socially connected (Pfeil et al., 2009). The analysis also showed that restaurant customers use a high rate of self-reference words (I, me, my) meaning the reviewers focus more on themselves and their emotions when representing themselves in social media websites. Additionally, younger people who are more open to using new technology and social media websites tend to more often spend their time online social media websites than older people (Hertzog & Huлтsch, 2000). Thus, the high usage of self-reference words in our result can be explained by the high percentile of younger people who leave their comments in online review website.

One important finding of the present study is that outdoor seating, delivery service, Wi-Fi, serving alcoholic beverages, presence of food for kids as in a children menu, accepting credit cards, plentiful parking spaces, good for groups and noise level were all shown to be significant factors affecting the overall ratings by restaurant reviewers. Similarly, Radojevic, and his colleagues (2014) find that the number of stars, room price, and the presence of air-conditioning, a lobby bar, and free Wi-Fi all positively affected customer satisfaction with a hotel. In addition, significant determinants of high ratings of restaurants depending on the average price of restaurants were identified by the current study. The average price was added in the analyses as an attempt to test the influence of restaurant type (e.g., quick-service, casual dining, fine dining) on significant determinants of overall ratings of online reviews. Although the price is not the only restaurant characteristic, the average price is a significant factor categorizing the type of restaurants. From a theoretical perspective, we extended Herzberg's Two Factor Theory in a restaurant context. Based on the theory, our findings revealed that certain characteristics of a restaurant are related to customer satisfaction, while different factors associated with restaurant dissatisfaction. Restaurants that provide outdoors seating, catering service, alcohol products, and private parking lot have higher ratings while the delivery service and good for dinner is found to lead to lower ratings. Thus, it is important for restaurant managers to fully acknowledge which service attributes are most likely to influence customers' satisfaction. A satisfied customer promotes positive WOM at no cost to the company and with credibility that are superior to those of conventional advertising (Yoo & Hanssens, 2008). In addition, depending on the type of restaurants, managers should

consider the potential influence of those offerings on their Yelp.com ratings. If restaurant owners hope to retain their competitive edge, they may need to offer significant accommodations to increase customer satisfaction. Overall, our results suggest that understanding the key attributes that amplify or dampen customer satisfaction can help restaurant managers or owners manage their service quality more effectively.

General implications

Theoretical implications

The present study is one of relatively few attempts to investigate the determinants of restaurants' ratings as used on social media websites. The uniqueness of the present study from an academic perspective is that it mixed qualitative and quantitative methods in gathering and analyzing data from a website of user-generated evaluative content (Yelp.com) to gain a better understanding of customer behavior in the area of online review writing and evaluative rating. Specifically, mixed quantitative and qualitative methods were used to identify factors affecting customers' satisfaction and comprehend customers' writing review behaviors by dividing into satisfier and dissatisfied in online review sites. While previous studies (Lundberg, Gudmundson, & Anderson, 2009; Sledge, Miles, & Coppage, 2008) have extensively applied Herzberg's Motivator and Hygiene Factor Theory to explain employees' job satisfaction in their workplaces, few studies have employed this theory in customers' restaurant brand satisfaction in online review sites. In addition, by taking into account factors affecting customer's satisfaction and dissatisfaction, our study will offer broader application of this theory to understand customer satisfaction in the context of social media. Furthermore, the findings of this study is to open up numerous avenues for future research on measuring customers' satisfaction at the online review sites.

Managerial implications

From a managerial perspective, the study provides restaurant owners and/or managers with insights into what their customers expect and how they perceive a restaurant. Customer comments on third-party online review websites can highlight areas needing improvement and enable restaurateurs to protect their brand online. Although some companies have their own social websites (e.g., blogs, comment pages, discussion forums, etc.), monitoring of comments appearing in third-party online review websites (e.g., Yelp) and a good working knowledge of the key attributes affecting restaurant customers' satisfaction can be the initial step when they plan service quality improvement endeavors. In addition, by identifying the most frequently appearing comments from satisfier or dissatisfier, it provides a frame of reference for online service providers to evaluate their service performance. Therefore, this study will provide a guidance for a company to strategically manage the online review site by acquiring a deep understanding of the phenomenon connected to online reviews.

Our findings suggest that restaurants should invest in learning what customers want to experience in the consumption context, what will involve them cognitively and emotionally in a manner leading to strong preference and loyalty toward the restaurant. In addition, it is important to give customers a pleasing shared experience, which is an important reason why people dine out. Ensuring that the meal experience along with providing high quality of service will result in positive comments in chat rooms, forums, and review websites. Thus,

considering the significant impact of pleasant service experience on customer satisfaction (Ali, Amin, & Cobanoglu, 2016), it is essential for hospitality companies to pay more attention to develop strategies on how a great experience and positive emotions affect customers' satisfaction or high rating in online review sites.

Our results can be an opportunity for hospitality practitioners to fine-tune their management strategies depending on the restaurants' different price points (low-priced vs. high-priced). Based on our findings, we advise that low-priced restaurants should focus on providing delivery service, good atmosphere for groups, outdoor seating, and parking space that lead to the customer satisfaction. On the other hand, in order to increase customer satisfaction for high priced restaurants, it is imperative to improve their food quality because of the value in foods that they customer in restaurants. This study would enable restaurant industry to have a better understanding of how they need to make a better allocation of resources and deliver the right offering, which lead to higher level of satisfaction by accommodating different customers' desire depending on restaurant price points.

Limitations and recommendations for future research

This study has several limitations that need to be considered. First, data analysis used only reviews on Yelp.com, which limits the study sample. It would be interesting to compare reviews of a particular restaurant on different websites. Second, the study design may have omitted some other possible factors. Future studies could expand the data set and include more possible variables (e.g., person-specific, location-specific, demographics) to assess their influence on customers' rating behaviors. Third, the results may not be generalizable since not all customers leave comments on online review websites. This may yield a sampling problem because customers who post reviews online might be different compared to the ones who do not post online reviews. Fourth, the influence of restaurant location on overall ratings of online reviews was not considered in the current study. The regional differences such as urban or isolated area as well as the number of competitors may be possible factors affecting overall ratings, which can be further investigated by future studies. Also, there is a particular set of users of the service which Yelp calls "elite" users who are a small group of in-the-know users and have a large impact on their local community. Since the current study did not differentiate such users in our analysis, future studies are advised to investigate the behavior and impact of such opinion leader users. The sample size presents another limitation as datamining practices would yield better results with larger samples. In our study, we only explored what factors affect customers' satisfaction in online review sites. Unfortunately, we did not directly investigated the relationship between WOM and customer's satisfaction, but it would be interesting to find condition or triggers leading satisfied customers to their WOM activity. Lastly, the current study did not propose directional hypotheses due to its explorative nature, which can be recommended for future research examining online reviews using quantitative approaches.

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